

Program : Diploma in Renewable Energy	
Course Code : 5293A	Course Title: Control Systems
Semester : 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L: 3 T: 1 P: 0)	Periods per semester: 60

Course Objectives:

- To provide a thorough knowledge about control engineering concepts.
- To provide an ability to comment on the stability of systems if the transfer functions are given.
- To lay the foundation for development of mathematical model for various systems.

Course Prerequisites:

Topic	Course Code	Course name	Semester
Basic knowledge of Electrical circuits		Fundamentals of Electrical and Electronics Engineering	1
Force and Torque		Applied Physics II 2	2
Basic knowledge in Mathematics		Engineering Mathematics I & II	1 & 2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Solve linear differential equations using Laplace transforms	12	Applying
CO2	Identify various representations of control systems	17	Applying

CO3	Develop various transient and steady state parameters of control systems	13	Applying
CO4	Identify different methods of determining stability of control systems	16	Applying
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Solve linear differential equations using Laplace transforms		
M1.01	Explain the concept of basic control systems	2	Understanding
M1.02	Classify different types of control systems.	2	Understanding
M1.03	Solve Laplace transform and inverse Laplace transform of common mathematical functions	4	Applying
M1.04	Apply Laplace transform to solve linear differential equations	4	Applying

Contents:

Basic block diagram and elements of control system - Control Systems-basic terms-plants-controller-feedback element-error detector- Basic structure(block diagram representations only) - meaning of physical systems, physical model and mathematical model (block diagram representations only)

Types of Control system- Linear time invariant and linear time variant system- open loop and closed loop controls system with physical examples.

Laplace Transform & Inverse Laplace Transform - Define Laplace transform- Laplace transforms of A, At, e^{at} , e^{-at} , Sin at, Cos at- Real differentiation, Real integration, Initial value and Final value theorems. Multiplication by e^{at} - Inverse Laplace transforms- Inverse Laplace transformation by partial fraction method (distinct and repeated roots only)

Solution of Linear differential equations - solution of linear differential equations -Simple problems (up to Second Order Systems)

CO2	Identify various representations of control systems		
M2.01	Develop transfer functions of control systems	4	Applying
M2.02	Solve poles and zeroes of a transfer function	2	Applying
M2.03	Construct block diagram of control systems	5	Applying
M2.04	Develop Signal flow graph of control systems	5	Applying
	Series Test – I	1	

Contents :

Transfer Functions - Transfer function- Definition- order of system- Procedure for determining transfer functions - Transfer function of simple electrical systems -RL circuit,- RC circuit-RLC circuit (derivations needed) - Advantages and disadvantages of transfer function-

Poles and zeros : Poles and zeros -basic concepts- determination of poles and zeros of simple transfer function- simple problems - graphical representation of poles and zeroes in S -plane

Block diagram - Block diagram -representation -rules for block diagram reduction - Determination of overall transfer function of simple control systems (single input-single output systems)- simple problems

Signal flow graph - Signal flow graph -Definition--representation of simple control system by signal flow graph -simple problems -various parameters of the signal flow graph- Rules for drawing signal flow graphs -Mason's gain formula (up to a maximum of 3 loops) - determination of the overall transmittance of control systems by signal flow graph (single input-single output systems).

CO3	Develop various transient and steady state parameters of control systems		
M3.01	Explain various responses, and standard test signals of control systems.	2	Understanding
M3.02	Illustrate various responses of a control system	3	Understanding
M3.03	Explain various time domain specifications	3	Understanding
M3.04	Identify various steady state parameters of control systems	5	Applying
Contents : Control system responses - Basic concept of transient and steady state responses - standard test signals and their transfer functions (Step, Ramp, Parabolic and Impulse signals) Transient response and Steady state response - Time response of first order system-Step input-Time constant-Time response of second order under damped system for step input Time Response specifications - Time domain specifications(definitions only) - Delay time - Rise time - Peak time - Peak Overshoot - Settling time (Relations & numerical problems based on Time domain specifications are not required) Steady State Error – Type number of control system - Steady state error - Static error coefficients - Static position, velocity, acceleration error constants - Steady state error in terms of K_p , K_v , K_a for Type 0, Type 1, Type 2 systems			
CO4	Identify different methods of determining stability of control systems		
M4.01	Utilize Routh Hurwitz criterion to determine the stability	5	Applying
M4.02	Construct Root locus to determine the stability	6	Applying
M4.03	Explain frequency response and frequency domain Specifications of control systems	2	Understanding
M4.03	Illustrate various control actions in industrial control systems	3	Understanding
	Series Test – II	1	
Contents : Stability of Control System: Definitions of Absolute and Relative stability - Routh Hurwitz criterion -statement-necessary and sufficient conditions - Solve simple problems using Routh Hurwitz criterion (up to 5 th order Normal Routh Array only)			

Root locus: Concept of root locus-Procedure to construct Root locus-Construct Root Locus for given transfer functions (Simple poles and zeros only)-stability from root loci

Frequency Response & Frequency Domain Specifications: Frequency response of control system - Definition-Frequency domain Specifications (definitions only)

Industrial control system -Block diagram-Control actions- definition -types of control actions -block diagram representation of On-Off, P, I, PI, PD and PID control actions

Text / Reference

T/R	Book Title/Author
T1	A Nagoor Kani, Control Systems, RBA Publications
T2	Nagarath I. J. and Gopal M., Control System Engineering, 5/e, New Age Publishers
T3	Ogata K, Modern Control Engineering, 5/e, Prentice Hall of India.
R1	Imthias Ahamed T. P, Control Systems, Phasor Books, 2016
R2	Gopal M., Control Systems Principles and Design, 4/e, Tata McGraw Hill.

Online Resources

Sl.No	Website Link
1	https://onlinecourses.nptel.ac.in/noc20_ee90/preview
2	https://www.youtube.com/playlist?list=PLYqSpQzTE6M8-wda5vbgHkMQTmu-21hRK
3	https://www.youtube.com/playlist?list=PLxn52v8fxX5l5tGzU1NAXRDkgqxK0k5UZ